

Humidity sensor reading prediction and incubator parameter determination

ABSTRACT:

This paper presents a proposed system designed for the development of a humidity sensor characterized by enhanced accuracy and reduced response time when compared to commercially available humidity sensors. The primary objective of the system is to ascertain the humidity within an incubator, wherein a preterm infant is housed. The core components of the system include the fabricated humidity sensor, a temperature sensor, and an incubator, collectively employed for continuous monitoring of pertinent parameters.

The research endeavours to employ machine learning algorithms to prognosticate optimal settings for incubator functionality, with an emphasis on achieving a high success rate in regulating temperature and humidity. Furthermore, the study extends to predicting the average values of breath rate and breath volume of the infant. Health factors such as the infant's weight in grams and gestational age are also considered in the predictive model.

The final predictive model developed and deployed had used linear regression which outperformed its counterparts ridge regression and decision tree models with a mean absolute error of 1.24. The experimental setup involves integrating the fabricated humidity sensor and a commercially available temperature sensor into an Arduino board to facilitate the determination of crucial parameters. This comprehensive approach seeks to enhance the efficiency of incubator systems for preterm infants through the integration of advanced sensing technologies and predictive modelling.

1.Introduction:

Preterm birth, commonly referred to as premature birth, denotes the delivery of an infant prior to reaching 38 weeks of gestational age, accompanied by a birth weight below two kilograms. This phenomenon signifies that the neonate has undergone an abbreviated period of intrauterine development and, consequently, is not fully equipped for independent survival. The truncated gestational period increases the likelihood of complex medical challenges, necessitating specialized care. Premature infants are particularly susceptible to a spectrum of health complications, with respiratory-related ailments ,pulmonary ailments being predominant. The underdeveloped nature of their lungs renders them more vulnerable to the exigencies of the external environment. This vulnerability underscores the critical need for meticulous monitoring and specialized interventions to address the unique medical exigencies associated with preterm birth.

In the year 2021, the incidence of preterm birth affected approximately one in every ten infants born in the United States, highlighting the prevalence and societal impact of this issue (source: Centers for Disease Control and Prevention, <https://www.cdc.gov/reproductivehealth/maternalinfanthealth/pretermbirth>). It is imperative to recognize that the prevalence of premature births surpasses general awareness, emphasizing the imperative for continued research, public health initiatives, and medical interventions to ameliorate the outcomes and well-being of preterm infants and their families.

2.Literature review:

There are various new methodologies for designing incubators recently. For example Dive and Kulkarni designed an incubator that can detect the light luminance inside the incubator and the audio voice of the premature baby in the incubator[1].The proposed incubator system can notify the doctor and nurse about the baby's condition. When the baby cries for any reason, the alarm is turned on and will go on until it is manually turned off.

In a study by Atul Kale ,They had used GSM technology to send alerts about the baby's condition.[2] . The Arduino board, in conjunction with a Sim 800 GSM Module as an information transmission submodule, delivers SMS to a medical expert the baby body's sensor data monitored by the system. In summary, this study provided new ways for designing a dependable baby hatchery using the UNO microcontroller.

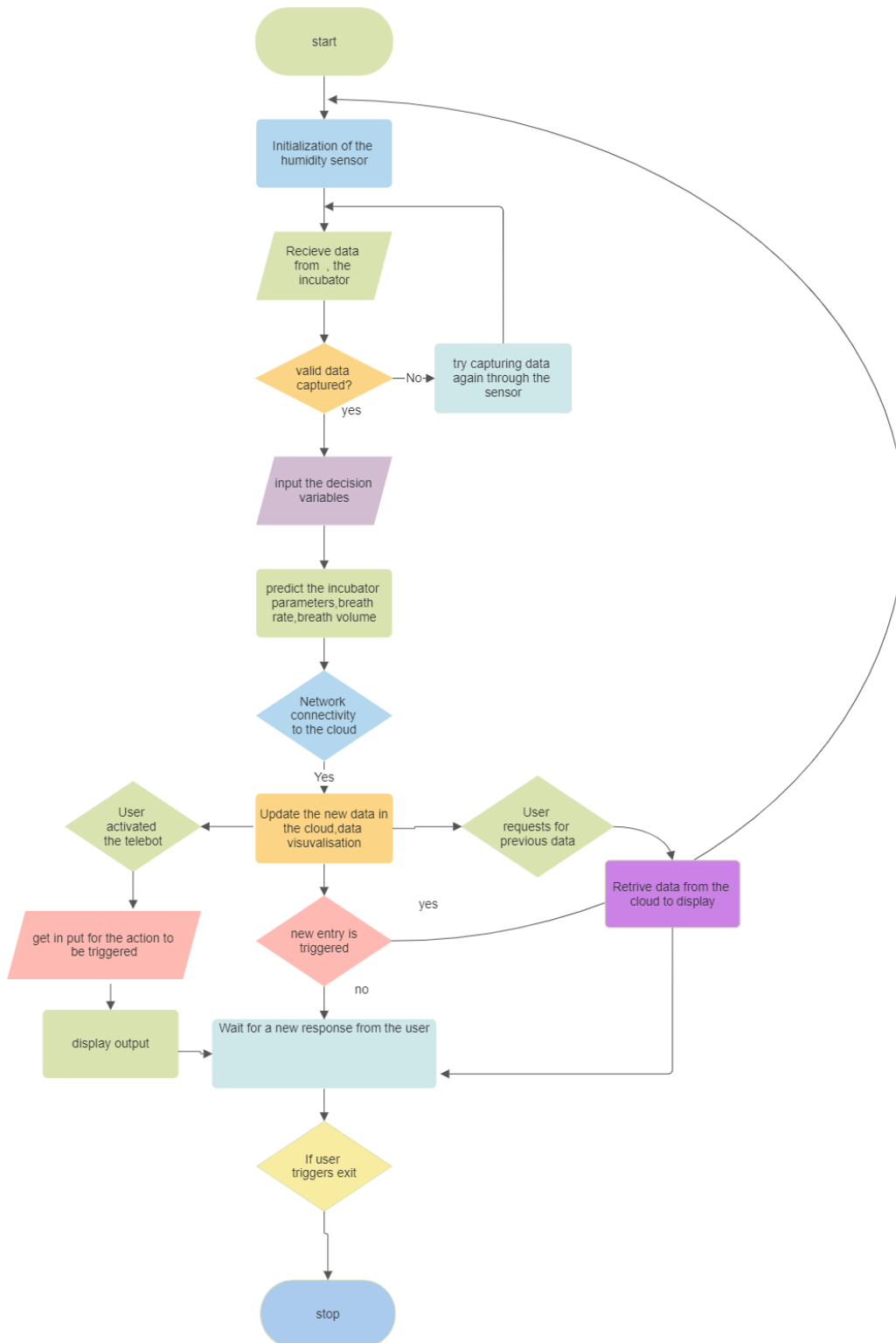
In the paper An android application based temperature and humidity monitoring and controlling child incubators paper by Mohaiminul Islam and Navila Rahman, preexisting dht11 had been used along with hc-06 and an interconnection of 2 incubators had been done wirelessly.(by means of the Bluetooth based android application)

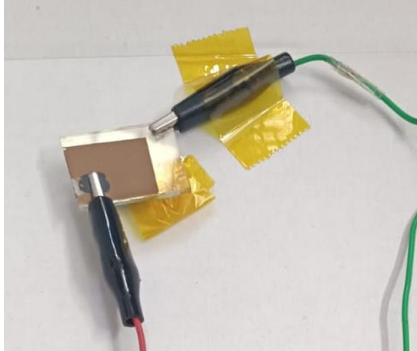
Non-contact physiological monitoring of preterm infants in the Neonatal Intensive Care Unit Mauricio Villarroel1*, Sitthichok Chaichulee 1 , João Jorge 1 , Sara Davis2 , Gabrielle Green2 , Carlos Arteta3 , Andrew Zisserman 3 , Kenny McCormick2 , Peter Watkinson4 and Lionel Tarassenko1 In this paper no sensors were used for the determination of parameters rather a contemporary approach of camera image capturing and use of image analysis models like cnn,naïve bayes were used to determine the parameters(multi-task Convolutional Neural Network (CNN), time periods during which the infant was present or absent from the incubator were automatically detected from the video recordings. Regions of interest (ROI) corresponding to skin were segmented from each video frame. These ROIs were used to extract cardiac-synchronous Photoplethysmographic Imaging (PPGi) and respiratory signals, from which heart rate and respiratory rate were estimated.)[11]

[12]Multiple Fault Detection and Smart Monitoring System Based on Machine Learning Classifiers for Infant Incubators Using Raspberry Pi 4(using a microprocessor) Maryam A. Mahdi*, Sabah A. Gittaffa, Abbas H. Issa - The proposed system had used four temperature sensors that were later extended to monitor skin temperature and two humidity sensors,The infant was monitored.

[13] Shabeeb et al. [12] proposed a method by which the humidity levels and air temperature in the infant's incubator are remotely monitored as a result of using an Arduino microcontroller with various sensors (DHT11/DHT22) and Internet of Things (IoT) applications. Using a wireless (ESP8266Wi-Fi) connection, the system connects to a network and is linked to a smartphone or computer application.

3.Workflow diagram:





3.Components used :

3.1 Fabricated humidity sensor:

Within the structure of the fabricated humidity sensor, the positive segment is comprised of PEDOT:PSS, serving as the p-type material of the electrical connection. Conversely, the glass substrate is coated with a mixture of ZnO and NiO, functioning as the N-type material and serving as the

negative terminal of the connection.

The escalation in humidity levels is denoted by an augmentation in oxygen ions, consequently resulting in heightened current conduction. Metal contacts are affixed to both terminals, thereby establishing a conductive pathway for electrical signals. This intricate arrangement facilitates the sensor's capacity to discern variations in humidity levels through the modulation of current conduction, offering a nuanced and responsive mechanism for humidity detection.

Fig 3.1.1 fabricated humidity sensor

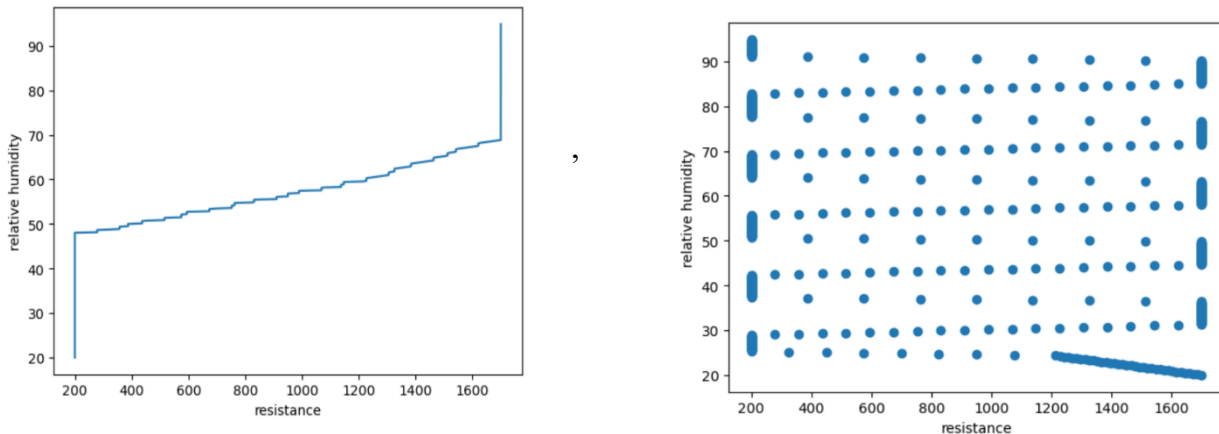


Fig 3.1.2: Graphical relation between resistance, relative humidity

By employing the aforementioned rationale, it becomes plausible to establish a quantitative relationship between the measured resistance in the humidity sensor and the corresponding relative humidity. The constructed models encompassed linear regression, ridge regression, and decision tree methodologies. Following a rigorous assessment, testing the linear regression

model exhibited superior performance, consequently being selected as the definitive model for deriving an expression correlating the resistance measurement with the ambient relative humidity. This selected linear regression model was subsequently utilized for the conclusive testing phase, affirming its efficacy in providing a robust and accurate representation of the intricate relationship between sensor resistance and ambient relative humidity levels of the surroundings. The use of machine learning techniques is justified as predicting an setting of optimum values of temperature and humidity by human medical practitioners may sometimes lead to faulty and unfavourable outcomes. Due to the training of the model with a variety of cases and their outcomes the setting of these parameters would be far more suitable and tailored for the infant inhabiting the incubator system. In the dataset a 20/80 ratio was used for test train split for checking the performance and accuracy of the developed model.

Confusion matrix for the various models built:

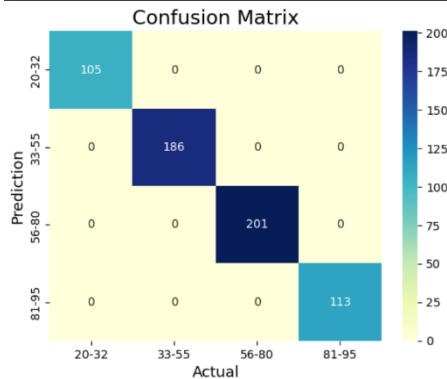


Fig 3.1.3 Confusion matrix linear regression

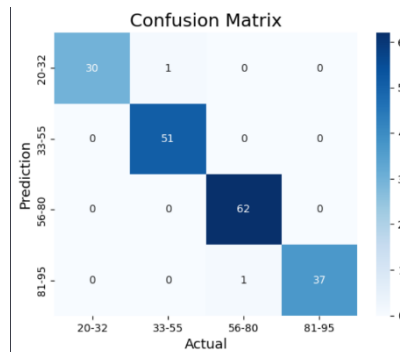


Fig 3.1.3 Cm decision tree

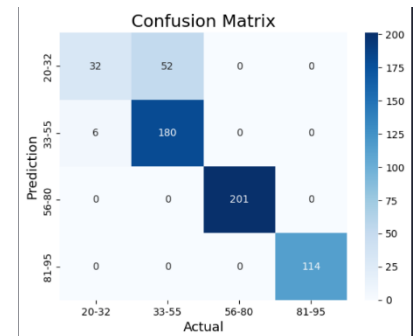


Fig 3.1.4 Confusion matrix ridge

Evaluation of the models built for the various devices:

Regression Model	Device Name														
	SD:CT-3					SD:CT-4					SD:CT-5				
	MSE	RMS E	MAE	R ²	Acc.	MSE	RM SE	MAE	R ²	Acc.	MSE	RM SE	MAE	R ²	Acc.
Decision Tree	15.5	3.93	0.41	0.99	0.98	15.5	3.93	0.41	0.99	0.98	15.5	3.93	0.41	0.99	0.98
Ridge Reg.	19.8	4.4	1.5	0.95	0.87	4.22	2.05	0.5	0.99	0.88	8.72	2.95	0.92	0.98	0.89
Linear Reg.	2.3 e ⁻²⁴	1.52 e ⁻¹²	1.24	0.99	1.0	2.1 e ⁻²⁴	1.4 e ⁻¹²	1.7 e ⁻¹²	0.99	0.99	2.4 e ⁻²⁴	1.5 e ⁻¹⁹	1.3 e ⁻¹²	0.99	1.0

Fig 3.1.2 The model evaluation for the various devices fabricated

1. precision	recall	f1-score	support	
	1	1.00	1.00	105
	2	1.00	1.00	186
	3	0.97	0.99	201
	4	1.00	1.00	113
2. accuracy			0.98	605
3. macro avg	0.98	0.99	1.00	605
4. weighted avg	1.00	1.00	1.00	605

Chart3.1.1 Classification report of linear regression model

1. precision	recall	f1-score	support	
i. 1	0.88	0.51	0.65	84
ii. 2	0.81	0.97	0.88	186
iii. 3	1.00	1.00	1.00	201
iv. 4	1.00	1.00	1.00	114
2. accuracy			0.92	585
3. macro avg	0.92	0.87	0.88	585
4. weighted avg	0.92	0.92	0.91	585

Chart3.1.2 Classification report of ridge regression model

i. precision	recall	f1-score	support	
b. 20-32	0.97	1.00	0.98	30
c. 32-55	1.00	0.98	0.99	52
d. 55-80	1.00	0.98	0.99	63
e. 80-95	0.97	1.00	0.99	37
2. accuracy			0.99	182
3. macro avg	0.99	0.99	0.99	182
4. weighted avg	0.99	0.99	0.99	182

Chart3.1.3 Classification report of decision tree model

The various parameters used to evaluate the model is as follows:

1. Accuracy

- **Definition:** Measure of the ratio of correct predictions to the total number of forecasts in model performance. A higher accuracy indicates a lower error rate.
- **Formula:** $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

2. Precision

- **Definition:** The proportion of true positive predictions relative to the sum of true positive and false-positive predictions. Indicates how many positive predictions were correct.
- **Formula:** $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

3. Recall

- **Definition:** The ratio of true positive predictions to the sum of true positive and false-negative predictions. Represents how many true positives were predicted or how many correct hits were found.
- **Formula:** $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

4. F1-Score

- **Definition:** A statistical metric for evaluating model performance using the harmonic mean of precision and recall.
- **Formula:** $\text{F1} = (2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Where:

- **TP (True Positive):** Predicted value is positive, and the actual one is positive.
- **TN (True Negative):** Predicted value is negative, and the actual one is negative.
- **FP (False Positive):** Predicted value is positive, but the actual one is negative.
- **FN (False Negative):** Predicted value is negative, but the actual one is positive.

These metrics collectively provide a comprehensive assessment of the model's performance, covering accuracy, precision, recall, and the F1-score.

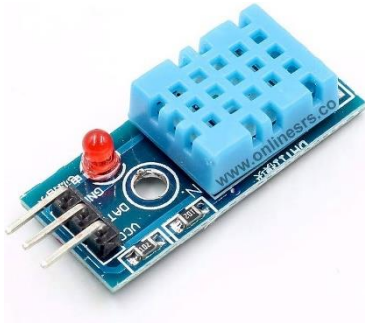


Fig 3.2.1 dht11

The DHT11 sensor utilizes a capacitive humidity sensor and a thermistor to gauge the ambient air's humidity and temperature, respectively.

The capacitive humidity sensor is responsible for determining the humidity level by measuring the changes in capacitance resulting from the absorption or release of water vapor by a hygroscopic material.

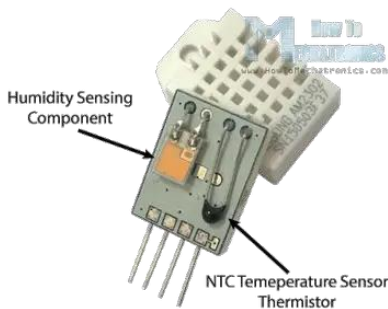


Fig 3.2.2 Internal structure of dht11

3.2 DHT11 Humidity sensor (humidity reference):

The DHT11 sensor, characterized by its economical nature, makes it a versatile and cost effective solution. It serves the purpose of assessing both humidity and temperature values. Within the experimental configuration, it was employed as a benchmark reference to validate the humidity measurements obtained from the fabricated humidity sensor and concurrently ascertain the internal temperature within the incubator.

Concurrently, a thermistor, a temperature-sensitive resistor, is employed to measure the temperature of the surroundings. The thermistor's resistance changes with variations in temperature, and this alteration is then utilized to derive the corresponding temperature value.

These internal components are integrated within the sensor's housing, creating a compact and efficient unit.

3.3 Arduino UNO Board:

Arduino is an open source computer hardware and software company single-board microcontrollers and microcontroller kits for building digital devices ,with all the various functionalities in built. Even when a microcontroller was not as capable as a microprocessor say for instance raspberry pi 4, The basic functionality and the control over devices was sufficient enough to make this project feasible, and interactive objects that can sense and control objects in the physical and digital world. Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards or Breadboards

(shields) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

4.Results and discussion:

The proposed infant incubator system:

In the proposed infant incubator system, the primary objective is to forecast and regulate crucial incubator parameters, specifically the temperature and relative humidity, to optimize the conditions for the well-being of the infant housed within the incubator. This predictive framework aims to determine the optimal setpoints for temperature and humidity, thereby maximizing the likelihood of favorable outcomes for the infant and to increase the chances of survival.

This conceptual framework can be extrapolated to hatcheries, broadening its applicability. By leveraging the predicted parameter values, the system ensures that the infant, or in the case of hatcheries, the developing organisms, inhabit an environment tailored for optimal growth and development. This predictive approach underscores a proactive strategy to maintain an ideal incubator environment, contributing to enhanced efficacy and success rates in the care of infants and hatchery operations

4.1The prediction of the incubator parameters are based on:

4.1.1 Age of the Infant: The age of the infant, quantified in gestational weeks, serves as a pivotal determinant of the infant's health status. A positive correlation exists between gestational age and the overall well-being of the infant, with a greater gestational age contributing to heightened health and increased chances of survival.

4.1.2 Weight of the Infant: The weight of the infant, measured in grams, functions as an approximate indicator of the infant's general growth and developmental status. This metric is instrumental in estimating the requisite parameters for the incubator, facilitating a nuanced approach to providing optimal conditions based on the infant's weight.

4.1.3 Room Temperature: The anticipated values of incubator parameters are subject to variation based on the ambient room temperature. Acknowledging and incorporating this external factor is essential for the accurate prediction and regulation of incubator conditions.

4.1.4 Room Humidity: Similar to room temperature, the projected values of incubator parameters are contingent upon the surrounding room humidity. Recognizing and accounting for this environmental variable is critical for refining the precision of the predicted incubator conditions.

4.1.5 Heat Out of Incubator: The heat dissipated during the operation of the incubator constitutes the heat outflow from the system. This factor is integral to comprehending the overall thermal dynamics of the incubator, thereby contributing to a comprehensive understanding of its operational characteristics.

In conjunction with the incubator parameters, the weight and gestational age of the infant are utilized to forecast the infant's breath rate and breath volume. This prediction is informed by reference to analogous cases involving infants exhibiting comparable developmental milestones, thereby enhancing the prognostic capabilities of the model.

4.2 The prediction of breath rate and breath volume is based on:

4.2.1 Age of the infant: The gestational period in weeks is the age of the infant , greater the gestational age ,greater is the health of the infant ,therefore the breath rate would be more sustained .With the predicted value of breath rate , in case of any variation in breath rate for the child, serious ailments can be identified at an earlier stage.

4.2.2.weight of the infant: The weight in grams of the infant is also a rough estimate of its overall growth and development, which can be used to estimate the breath rate and the breath volume.

As mentioned previously, the humidity inside the incubator is measured using the fabricated sensor and the temperature using dht11, therefore the parameters currently present inside the incubator are measured and displayed and the predicted values are also displayed, and immediately stored to the cloud. The system is also connected to a telegram bot ,so even when the medicinal practitioner is at a far off location, they would have access to the data through

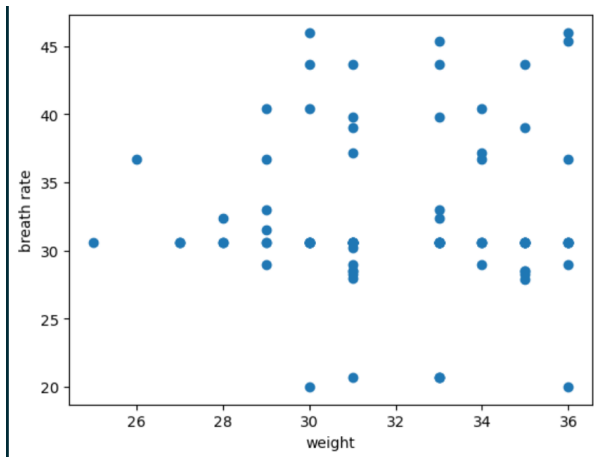


Fig 4.2.1 Relation between weight and breath rate

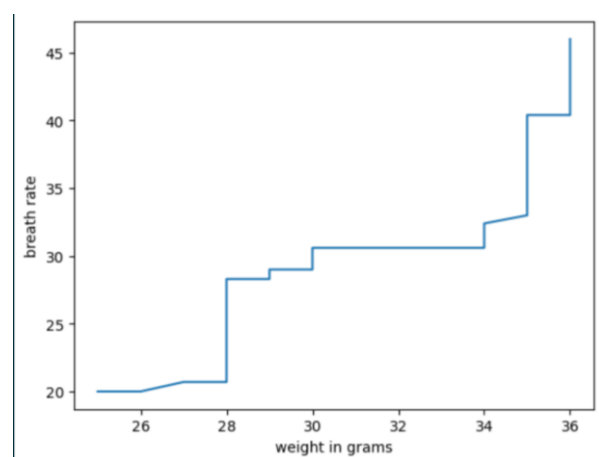


Fig 4.2.1 Relation between weight and breath rate

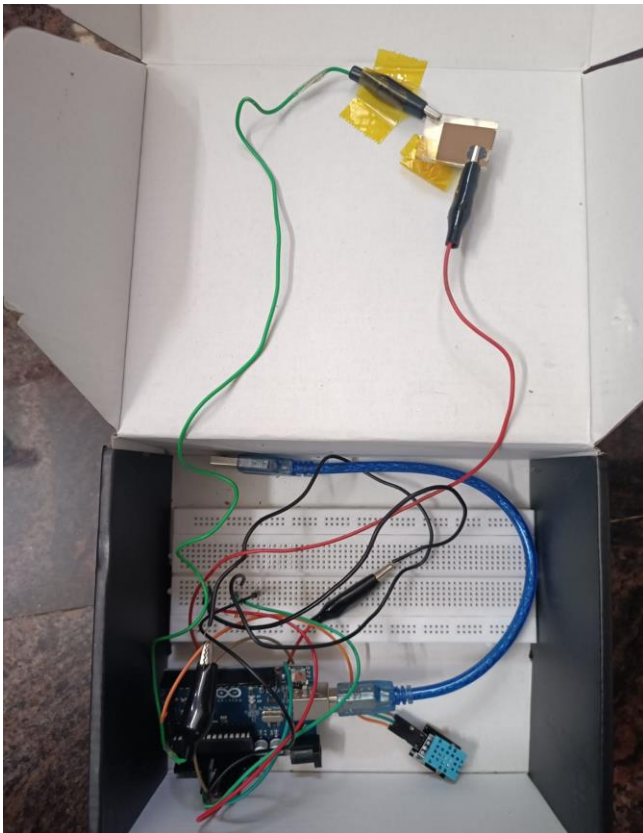
the telegram bot or through the cloud directly.

The incubator parameters which should be set for a particular child by taking the weight,age,room temperature,room humidity .

Model Evaluation:

Regression Model	Value measured									
	Breath rate					Breath volume				
	MSE	RM SE	MAE	R ²	Acc.	MSE	RM SE	MAE	R ²	Acc.
Linear Reg.	30.3367	5.50788	3.89	0.73	0.94	48.6866	6.977	48.42	0.92	0.80

4.2:The external circuit connected:



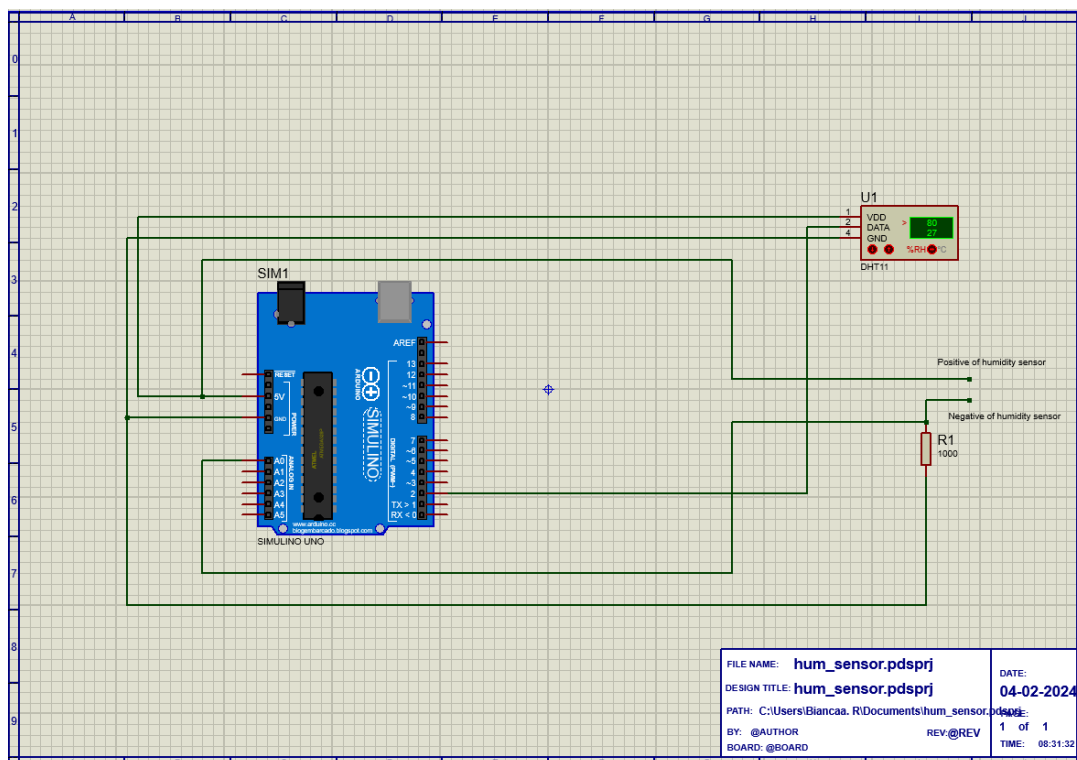
The schematic representation of the external circuit is depicted, wherein the Arduino unit is energized through connection to a laptop. The DHT11 sensor is integrated and energized via the 3.3V power source. A stable 5V is supplied as input to the reference resistance and the fabricated humidity sensor combination. The analog pin A0 is designated for reading the voltage differential across the fabricated humidity sensor. The resultant value is within the range of 0 to 1023 and is subsequently converted from a 5V scale.

The preference for the use of Arduino instead of other edge devices would be that, since it's a cost effective microcontroller it is more suitable for this purpose and employing a microprocessor like raspberry pi would be unnecessary for the tiniest of computational tasks.

Fig 4.2.1 External circuit

The module PyQt5 used for GUI is more suitable to be worked with Arduino as it could continue running in the host system, on the other hand if it was enabled in a raspberry pi it would have been run on Debian os which has certain GUI functionalities different. The analog to digital pin available in Arduino was another reason to gravitate towards it as there would not be a need to establish an external analog to digital converting circuit.\

Predicated upon the discerned voltage drop across the sensor and the drop across the established reference resistance, the resistance of the humidity sensor can be approximated. This derived resistance value is then input into the machine learning model to facilitate predictions pertaining to the measured relative humidity. This comprehensive circuit configuration enables the acquisition of precise sensor data, subsequently utilized for predictive modeling of environmental humidity levels.



4.2.2 Circuit diagram in proteus

4.2.2 Calculation involved:

```
sensorValue = analogRead(sensorPin); // Read Vout on analog input pin A0
(Arduino can sense from 0-1023, 1023 is 5V)
Vout = (Vin * sensorValue) / 1023; // Convert Vout to volts
```

$$R = (((V_{in} * R_{ref}) / V_{out}) - R_{ref});$$

Where V ref is 1000 ohms (Taken in accordance to the dataset of the sensor)

Therefore by multiplying the analog output value's fraction with 5v a voltage drop in the range (0-5 v) is found, Using which the current value of attained resistance is obtained.

4.3 Application Development:

The developed application is a desktop application created entirely using the Python programming language, with the integration of the PyQt5 module for the graphical user interface (GUI) and MySQL for database interaction.

4.3.1 Features in the Application:

1. **Record Register:** The application incorporates a record register, which archives and retrieves all historical data from the cloud. This feature ensures accessibility to previously added and stored data through the application.
2. **Authentication System:** A secure signup and login mechanism is implemented, integrating with a database to restrict unauthorized access and safeguard sensitive information.
3. **Real-time Prediction:** The application provides a functionality to add new records and receive immediate prediction results. The machine learning model operates in the backend, contributing to real-time data analysis.
4. **Customized Dashboard:** Each user is endowed with a personalized dashboard tailored to individual preferences and requirements, enhancing user experience and facilitating efficient data management.
5. **Telegram Bot Integration:** The application seamlessly integrates with a Telegram bot, allowing convenient data access. Medical professionals, such as doctors, can effortlessly retrieve data via the Telegram bot by submitting requests.
6. **Cloud Integration (ThingSpeak):** The application ensures seamless and immediate transmission of added values to the cloud. The chosen cloud service for this purpose is ThingSpeak, facilitating robust and glitch-free data storage and retrieval. This cloud integration enhances data accessibility and facilitates remote monitoring.

4.3.2: The user interface:

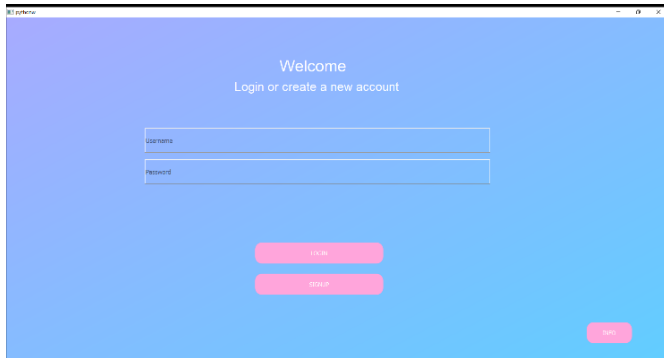


Fig 4.3.2.1 login screen

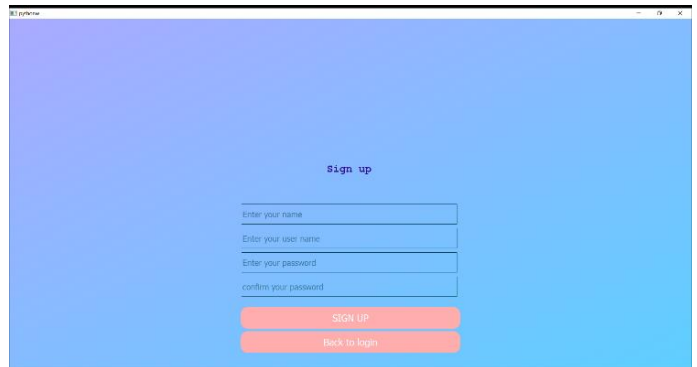


Fig 4.3.2.2 signup screen

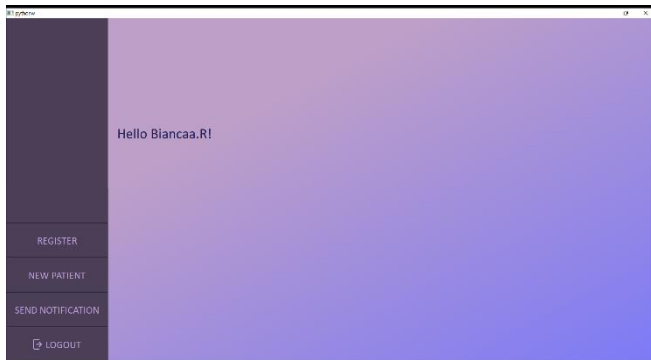


Fig 4.3.2.3 customized dashboard



Fig 4.3.2.4 Older records

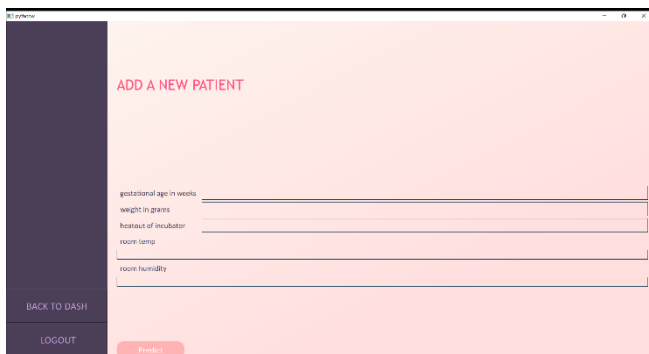


Fig 4.3.2.5 Adding new inputs



Fig 4.3.2.6 Telepage for activating the telegrambot



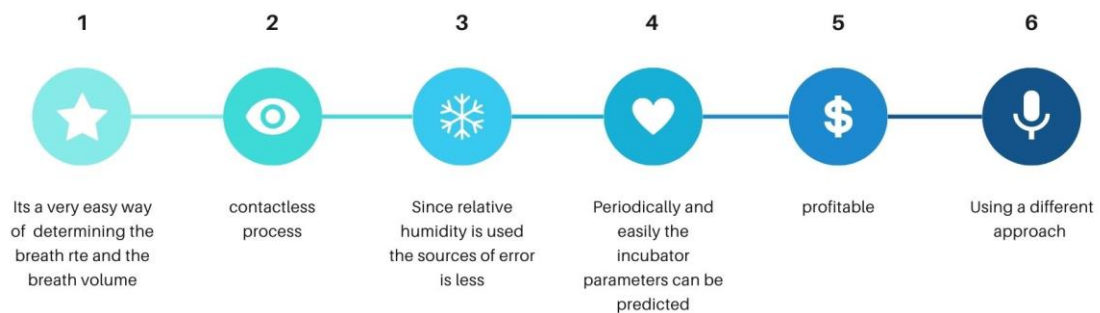
Fig4.3.2.7 Prototype of the proposed system

5:APPLICATION:

The application of this study is that it can be used in hospitals and if used in a large scale, a great number of premature babies, who die each year can be saved.

ADVANTAGES OF THE STUDY

Why to use learnbreathe?



6.CONCLUSION:

In conclusion, this study was undertaken to examine the conditions surrounding preterm infants and propose strategies to enhance their longevity by predicting optimal incubator settings, specifically temperature and humidity, to maximize their likelihood of survival. The prediction model incorporates breath rate and breath volume data from infants with similar developmental profiles, facilitating the identification and analysis of any deviations in the respiratory patterns of a given child. The overarching objective of this study is to mitigate the incidence of mortality attributed to preterm births.

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- [11] Non-contact physiological monitoring of preterm infants in the Neonatal Intensive Care Unit Mauricio Villarroel^{1*}, Sitthichok Chaichulee¹, João Jorge¹, Sara Davis², Gabrielle Green², Carlos Arteta³, Andrew Zisserman³, Kenny McCormick², Peter Watkinson⁴ and Lionel Tarassenko

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